Lab 4 Submission

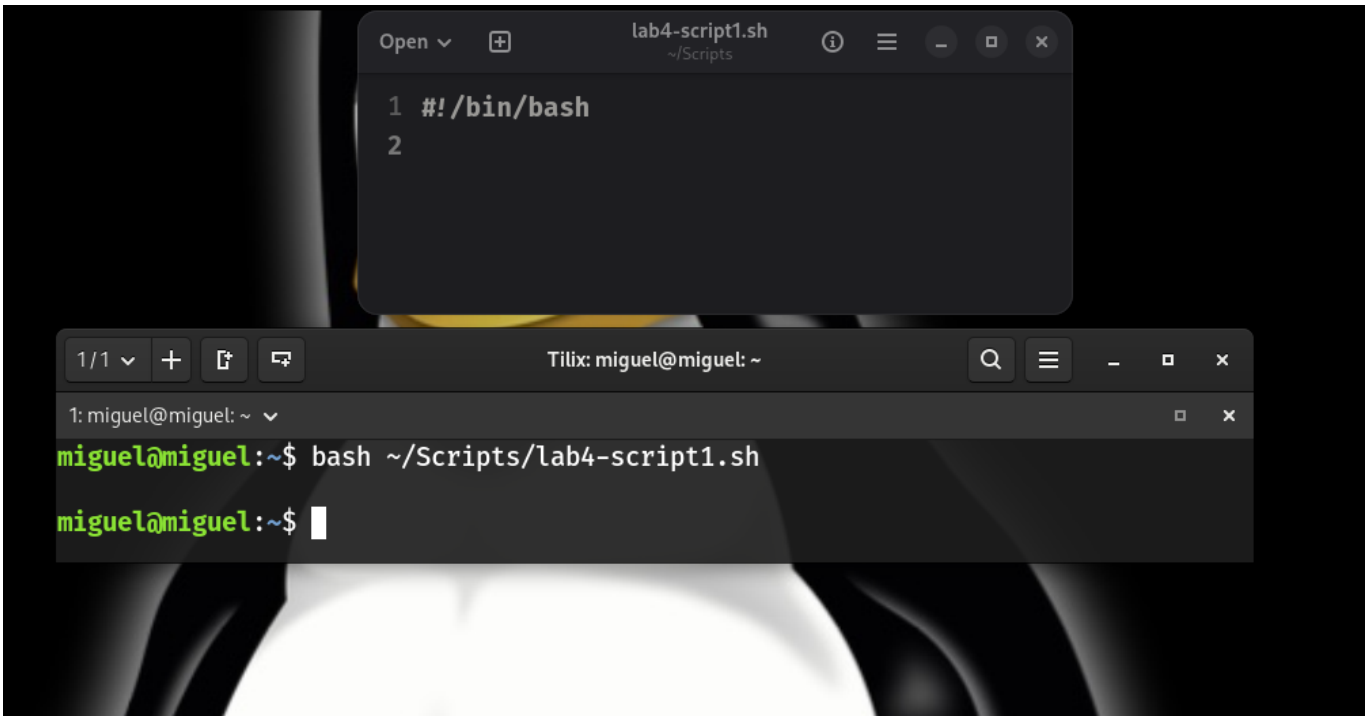
Question 1

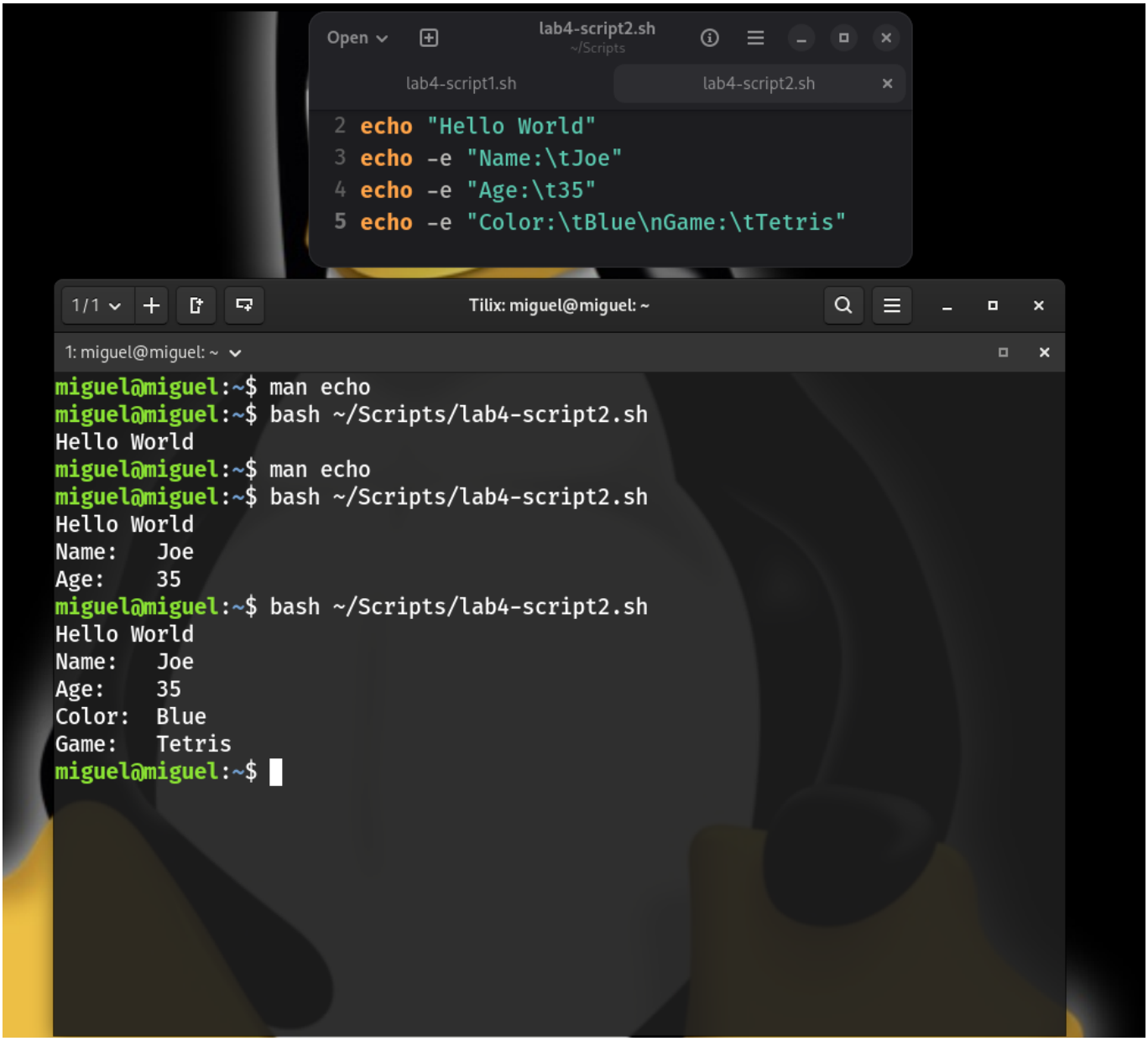
```
1/1                                   Tilix: miguel@miguel: ~
1: miguel@miguel: ~                   3: miguel@miguel: ~
bastet/stable 0.43.2-2 amd64          miguel@miguel:~$ apt search --names-only "tetris"
  ncurses Tetris clone with a bastard algorithm   games-tetris/stable 5 all
                                        Debian's tetris-like games
ltris/stable 2.0.3-1 amd64            vitetris/stable 0.59.1-3 amd64
  very polished Tetris clone with CPU opponents     Virtual terminal *tris clone
tint/stable 0.07 amd64                miguel@miguel:~$
  Tetris clone for text based terminal
miguel@miguel:~$
2: miguel@miguel: ~
bastet/stable 0.43.2-2 amd64
  ncurses Tetris clone with a bastard algorithm

blockattack/stable 2.9.0-1+b1 amd64
  puzzle game inspired by Tetris

blockout2/stable 2.5+dfsg1-2 amd64
  Tetris like game (3D-tetris)

blocks-of-the-undead/stable 1.0-7 amd64
lines 1-10
```

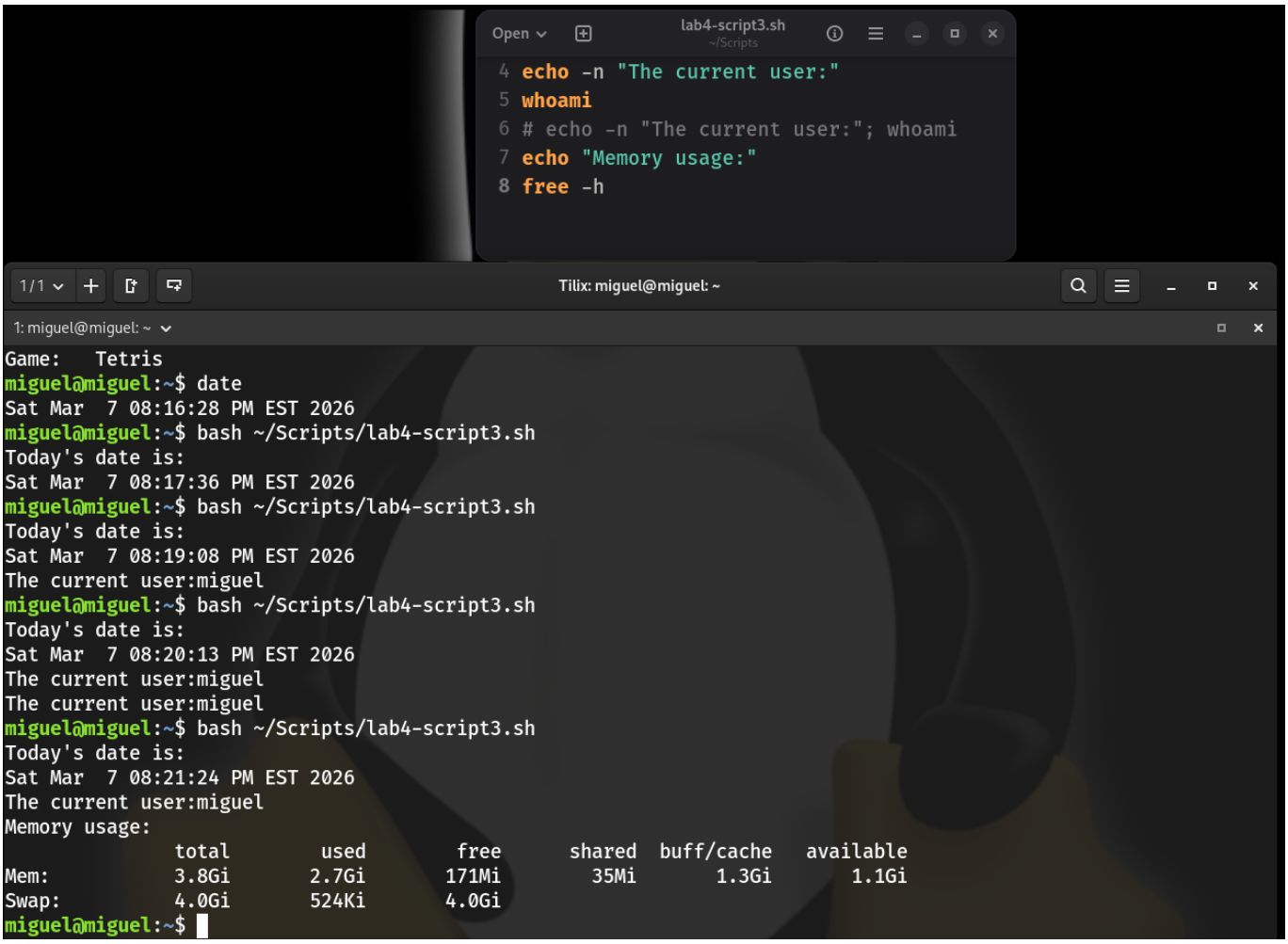
```
1/1                                   Tilix: miguel@miguel: ~
1: miguel@miguel: ~
miguel@miguel:~$ sudo apt install blockout2+ cuyo+ blockattack-
[sudo] password for miguel:
The following packages were automatically installed and are no longer required:
  libfmt10      libphysfs1          libsdl2-mixer-2.0-0  libxmp4
  libopusfile0  libsdl2-image-2.0-0  libsdl2-ttf-2.0-0
Use 'sudo apt autoremove' to remove them.

Installing:
  blockout2  cuyo

Installing dependencies:
  cuyo-data  libmikmod3  libsdl-mixer1.2  libsdl1.2debian

REMOVING:
  blockattack

Summary:
  Upgrading: 0, Installing: 6, Removing: 1, Not Upgrading: 4
  Download size: 8,078 kB
  Space needed: 4,102 kB / 234 GB available

Continue? [Y/n] Y
Get:1 http://deb.debian.org/debian trixie/main amd64 libmikmod3 amd64 3.3.13-1 [135 kB]
Get:2 http://deb.debian.org/debian trixie/main amd64 libsdl1.2debian amd64 1.2.68-3 [85.9 kB]
Get:3 http://deb.debian.org/debian trixie/main amd64 libsdl-mixer1.2 amd64 1.2.12-19+b1 [81.6 kB]
Get:4 http://deb.debian.org/debian trixie/main amd64 blockout2 amd64 2.5+dfsg1-2 [4,228 kB]
Get:5 http://deb.debian.org/debian trixie/main amd64 cuyo-data all 2.1.0-1 [3,372 kB]
Get:6 http://deb.debian.org/debian trixie/main amd64 cuyo amd64 2.1.0-1 [175 kB]
Fetched 8,078 kB in 0s (24.7 MB/s)
(Reading database ... 233728 files and directories currently installed.)
Removing blockattack (2.9.0-1+b1) ...
Selecting previously unselected package libmikmod3:amd64.
(Reading database ... 233701 files and directories currently installed.)
Preparing to unpack .../0-libmikmod3_3.3.13-1_amd64.deb ...
```

```
miguel@miguel:~$ sudo apt remove cuyo
The following packages were automatically installed and are no longer required:
  cuyo-data  libopusfile0  libsdl2-image-2.0-0  libsdl2-ttf-2.0-0
  libfmt10   libphysfs1    libsdl2-mixer-2.0-0  libxmp4
Use 'sudo apt autoremove' to remove them.

REMOVING:
  cuyo

Summary:
  Upgrading: 0, Installing: 0, Removing: 1, Not Upgrading: 4
  Freed space: 489 kB

Continue? [Y/n] Y
(Reading database ... 234798 files and directories currently installed.)
Removing cuyo (2.1.0-1) ...
Processing triggers for hicolor-icon-theme (0.18-2) ...
Processing triggers for gnome-menus (3.36.0-3) ...
Processing triggers for mailcap (3.74) ...
Processing triggers for desktop-file-utils (0.28-1) ...
miguel@miguel:~$ sudo apt purge cuyo
The following packages were automatically installed and are no longer required:
  cuyo-data  libopusfile0  libsdl2-image-2.0-0  libsdl2-ttf-2.0-0
  libfmt10   libphysfs1    libsdl2-mixer-2.0-0  libxmp4
Use 'sudo apt autoremove' to remove them.

REMOVING:
  cuyo*

Summary:
  Upgrading: 0, Installing: 0, Removing: 1, Not Upgrading: 4
  Space needed: 0 B / 234 GB available

Continue? [Y/n] Y
```

Question 2


Open
lab4-script1.sh
~/Scripts
1 #!/bin/bash
2
1/1
Tilix: miguel@miguel: ~
1: miguel@miguel: ~
miguel@miguel:~$ bash ~/Scripts/lab4-script1.sh
miguel@miguel:~$


Open
lab4-script2.sh
~/Scripts
lab4-script1.sh
lab4-script2.sh
2 echo "Hello World"
3 echo -e "Name:\tJoe"
4 echo -e "Age:\t35"
5 echo -e "Color:\tBlue\nGame:\tTetris"
1/1
Tilix: miguel@miguel: ~
1: miguel@miguel: ~
miguel@miguel:~$ man echo
miguel@miguel:~$ bash ~/Scripts/lab4-script2.sh
Hello World
miguel@miguel:~$ man echo
miguel@miguel:~$ bash ~/Scripts/lab4-script2.sh
Hello World
Name: Joe
Age: 35
miguel@miguel:~$ bash ~/Scripts/lab4-script2.sh
Hello World
Name: Joe
Age: 35
Color: Blue
Game: Tetris
miguel@miguel:~$


Open
lab4-script3.sh
~/Scripts
4 echo -n "The current user:"
5 whoami
6 # echo -n "The current user:"; whoami
7 echo "Memory usage:"
8 free -h
1/1
Tilix: miguel@miguel: ~
1: miguel@miguel: ~
Game: Tetris
miguel@miguel:~$ date
Sat Mar 7 08:16:28 PM EST 2026
miguel@miguel:~$ bash ~/Scripts/lab4-script3.sh
Today's date is:
Sat Mar 7 08:17:36 PM EST 2026
miguel@miguel:~$ bash ~/Scripts/lab4-script3.sh
Today's date is:
Sat Mar 7 08:19:08 PM EST 2026
The current user:miguel
miguel@miguel:~$ bash ~/Scripts/lab4-script3.sh
Today's date is:
Sat Mar 7 08:20:13 PM EST 2026
The current user:miguel
The current user:miguel
miguel@miguel:~$ bash ~/Scripts/lab4-script3.sh
Today's date is:
Sat Mar 7 08:21:24 PM EST 2026
The current user:miguel
Memory usage:
total used free shared buff/cache available
Mem: 3.8Gi 2.7Gi 171Mi 35Mi 1.3Gi 1.1Gi
Swap: 4.0Gi 524Ki 4.0Gi
miguel@miguel:~$

Question 3

challenge_lab4.sh
~/cis106/labs/lab4

```bash
#!/bin/bash
length=100

for ((i=0; i<length; i++)); do
    printf "-"
done
printf "\n"


figlet "Hostname:"
hostname


figlet "System:"
uname

echo "DATE EST: "
date +"%D %T"


figlet "Uptime:"
uptime


echo "Memory Usage:"
free -h


figlet "Disk Usage:"
df -h

for ((i=0;i<100;i++)); do
    printf "-"
done
printf "\n"


```

Tilix: miguel@miguel: ~

```
tmpfs           393M  164K  393M   1% /run/user/1000
----------------------------------------------------------------------------------------------------
miguel@miguel:~$ bash ~/cis106/labs/lab4/challenge_lab4.sh
----------------------------------------------------------------------------------------------------
 _   _          _
| | | | ___  ___| |_ _ __   __ _ _ __ ___   ___ _
| |_| |/ _ \/ __| __| '_ \ / _` | '_ ` _ \ / _ (_)
|  _  | (_) \__ \ |_| | | | (_| | | | | | |  __/_
|_| |_|\___/|___/\__|_| |_|\__,_|_| |_| |_|\___(_)

miguel
 ____           _
/ ___| _   _ ___| |_ ___ _ __ ___  _
\___ \| | | / __| __/ _ \ '_ ` _ \(_)
 ___) | |_| \__ \ ||  __/ | | | | |_
|____/ \__, |___/\__\___|_| |_| |_(_)
       |___/
Linux
DATE EST:
03/08/26 07:10:31
 _   _      _  _
| | | |    | |(_)
```

```
| | | | '_ \| __| | '_ ` _ \ / _ (_)
| |_| | |_) | |_| | | | | | |  __/_
 \___/| .__/ \__|_|_| |_| |_|\___(_)
      |_|
 07:10:31 up  1:53,  1 user,  load average: 0.25, 0.16, 0.12
Memory Usage:
              total        used        free      shared  buff/cache   available
Mem:          3.8Gi       2.0Gi       600Mi        56Mi       1.5Gi       1.8Gi
Swap:         4.0Gi          0B       4.0Gi

 ____  _     _     _  _
|  _ \(_)___| | __ | | |___  __ _  __ _  ___ _
| | | | / __| |/ / | | | / __|/ _` |/ _` |/ _ (_)
| |_| | \__ \   <  | |_| \__ \ (_| | (_| |  __/_
|____/|_|___/_|\_\  \___/|___/\__,_|\__, |\___(_)
                                     |___/
Filesystem      Size  Used Avail Use% Mounted on
udev            1.9G     0  1.9G   0% /dev
tmpfs           393M  4.4M  388M   2% /run
/dev/sda1       242G   11G  219G   5% /
tmpfs           2.0G   12K  2.0G   1% /dev/shm
tmpfs           5.0M  8.0K  5.0M   1% /run/lock
tmpfs           1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
tmpfs           2.0G   16K  2.0G   1% /tmp
/dev/loop1       15M   15M     0 100% /snap/cheat/4279
/dev/loop2       64M   64M     0 100% /snap/core20/2717
/dev/loop0       49M   49M     0 100% /snap/snapd/25935
tmpfs           393M  164K  393M   1% /run/user/1000
-----------------------------------------------------------------------------------------------------
miguel@miguel:~$
```

script